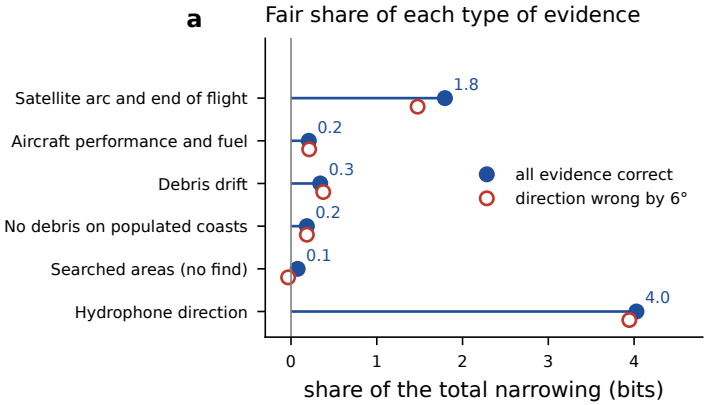
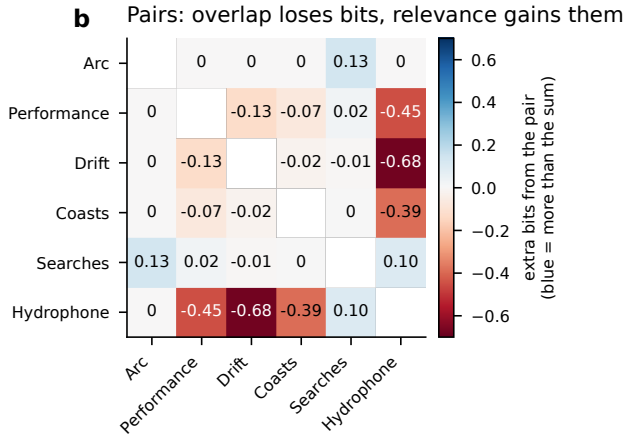
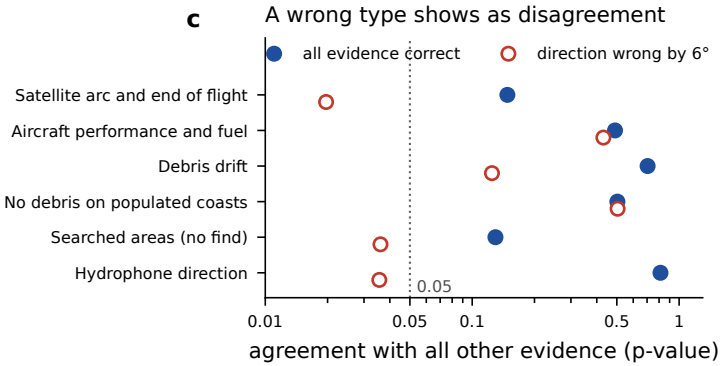
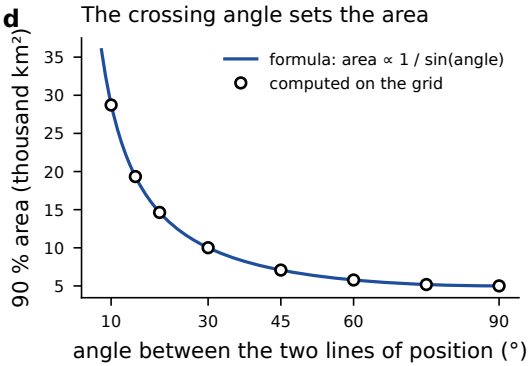
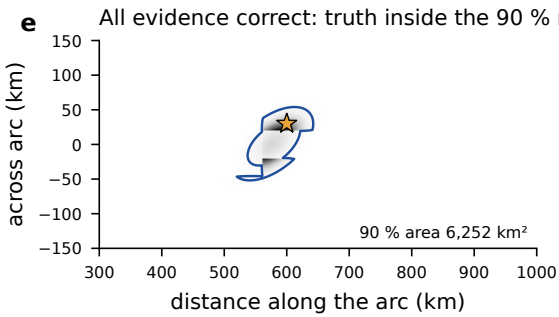
How much each type of evidence narrows the area, and a test that the types agree

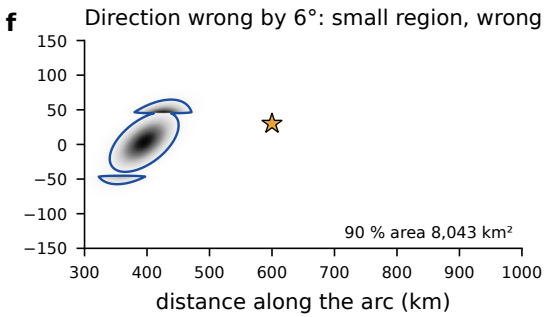












Note (STE100): This is a teaching example. All numbers are invented. It is not an MH370 result. (a) How many bits each type of evidence gives, averaged over every order in which the types can be added. The shares add to the total. One bit halves the area. (b) Bits from a pair, minus the bits of each type alone. Red: the two types cut the same area. Blue: one type makes the other more useful. Near zero: the two cuts are independent, so the area reductions multiply. (c) A test of each type against all the others. A low value means that the evidence does not agree. (d) As in a navigation fix, two lines of position that cross at a small angle leave a large area. (e, f) The final map when all evidence is correct, and when the hydrophone direction is wrong by 6 degrees.
Technical: same invented model as the first figure. (a) Shapley values of the information gain over all 64 subsets. (b) I(A,B) - I(A) - I(B), in bits. (c) Suspiciousness ln S = ln R - ln I (Handley and Lemos 2019) for each type against the product of the other five, flat prior on the grid; p-value from chi-squared with 2 degrees of freedom (Gaussian calibration, approximate here). (d) Two Gaussian ridges, sd 20 and 17 km; formula A90 = 2 pi ln(10) s1 s2 / sin(angle). (e, f) The 6 deg error is 157 km at 1,500 km range.